

Complex Systems Exercise 3

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1 First passage time in 1D

1.1 Motivation

Many systems can be model using FPT. Here are a few non-physical examples:

- The time between an even occouring and it being on the news.
To formalize this, we can think of society as a lattice, where every person is a site with a different number of neighbors based on different forms of communication. Concepts and pieces of information act as particles moving in this lattice. This way, the knowledge of an event happning will spread through this system according to a diffusion-like equation (not the diffusion equation itself since the total amount of knowledge is not conserved). So, we can use FPT to find the time it will take the knowledge of an event to reach a reporter, and assuming a preknown distribution of the "reporter to article time", we can find the time it will take it to reach the news.
- The time it will take a stock to pass a limit.
Stocks are many times bought and sold once they reach a certain price. Since, as we discussed, stocks can be modeled using the diffusion equation, we can find the average time to reach such a limit. This is helpful when choosing the limit int he first place, and when selecting the cycle time for the algorithm.
- The time for the signal to pass in a synapses between neurons.
The neural paths are constructed by a linear connection of neurons, each recieving information from the dendrites on one side and passing it on to the axon on the other side. The axon is then connected to other neurons via synapses - small gaps in which neurotransmitters live. When the axon become electrically charged, it releases the neurotransmitters. They then

diffuse through the synapse, and reach the dendrites of the next neuron in line. The time it will take from one neuron releasing neurotransmitters until they reach the next neuron can be found using FPT. The passage inside the neuron is a result of a repeating chain of chemical reactions, the time of which can also be calculated. Thus, we can get the total time it would take for a signal to pass along a nerve.

1.2 Equation and boundary conditions

Since we deal with a diffusion problem in 1D, the appropriate PDE is simply the diffusion equation:

$$\partial_t P_{x_0} = -D\partial_x^2 P_{x_0} \quad (1)$$

For the boundary condition, we must require that the particle has probability zero to have reached x_0 , at any time t . This will allow us later to find the time t for any given x_0 :

$$P_{x_0}(x_0, t) = 0 \quad (2)$$

1.3 Solving the equation

Starting from a point source

$$P_{x_0}(x, t = 0) = \delta(x) \quad (3)$$

And using the analogy to electrostatics to create an image source at $x = 2x_0$ with negative value:

$$P_{x_0}(x, t = 0) = \delta(x) - \delta(x - 2x_0) \quad (4)$$

Since the Green's function of the diffusion equation is the gaussian with standard deviation $\sqrt{2Dt}$, and since it's a linear equation, we get:

$$P_{x_0}(x, t) = \frac{1}{\sqrt{4\pi Dt}} \left(\exp\left[-\frac{x^2}{2Dt}\right] - \exp\left[-\frac{(x - 2x_0)^2}{2Dt}\right] \right) \quad (5)$$

Checking the boundary condition is satisfied:

$$\begin{aligned} P_{x_0}(x_0, t) &= \frac{1}{\sqrt{4\pi Dt}} \left(\exp\left[-\frac{x_0^2}{2Dt}\right] - \exp\left[-\frac{(x_0 - 2x_0)^2}{2Dt}\right] \right) = \\ &= \frac{1}{\sqrt{4\pi Dt}} \left(\exp\left[-\frac{x_0^2}{2Dt}\right] - \exp\left[-\frac{x_0^2}{2Dt}\right] \right) = 0 \end{aligned} \quad (6)$$

As required.

1.4 Survival probability

We are looking for the probability for the particle to be in one of the states described by P_{x_0} , therefore, we must sum over all $x < x_0$.

$$\begin{aligned} S(t) &= \int_{-\infty}^{x_0} P_{x_0}(x, t) dx = \\ &= \frac{1}{\sqrt{4\pi Dt}} \left(\int_{-\infty}^{x_0} \exp\left[-\frac{x^2}{2Dt}\right] dx - \int_{-\infty}^{x_0} \exp\left[-\frac{(x-2x_0)^2}{2Dt}\right] dx \right) \end{aligned} \quad (7)$$

The first integral gives:

$$\begin{aligned} &\int_{-\infty}^{x_0} \exp\left[-\frac{x^2}{2Dt}\right] dx = \\ &= \int_{-\infty}^0 \exp\left[-\frac{x^2}{2Dt}\right] dx + \int_0^{x_0} \exp\left[-\frac{x^2}{2Dt}\right] dx = \\ &= \sqrt{\pi Dt} + \sqrt{\pi Dt} \operatorname{erf}\left(\frac{x_0}{\sqrt{4Dt}}\right) \end{aligned} \quad (8)$$

Where erf is the error function

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (9)$$

Similarly, the second integral:

$$\begin{aligned} \int_{-\infty}^{x_0} \exp\left[-\frac{(x-2x_0)^2}{2Dt}\right] dx &= \int_{-\infty}^{-x_0} \exp\left[-\frac{x^2}{2Dt}\right] dx = \int_{x_0}^{\infty} \exp\left[-\frac{x^2}{2Dt}\right] dx = \\ &= \sqrt{\pi Dt} - \sqrt{\pi Dt} \operatorname{erf}\left(\frac{x_0}{\sqrt{4Dt}}\right) \end{aligned} \quad (10)$$

And in total, we get:

$$S(t) = \frac{1}{\sqrt{4\pi Dt}} \left(\sqrt{\pi Dt} + \sqrt{\pi Dt} \operatorname{erf}\left(\frac{x_0}{\sqrt{4Dt}}\right) - \sqrt{\pi Dt} + \sqrt{\pi Dt} \operatorname{erf}\left(\frac{x_0}{\sqrt{4Dt}}\right) \right) \quad (11)$$

$$S(t) = \operatorname{erf}\left(\frac{x_0}{\sqrt{4Dt}}\right) \quad (12)$$

1.5 The First Passage Time distribution

We need to find the probability that time t is the first time where the particle reached x_0 . Since the survival probability shows the total probability that the

particle didn't get to x_0 up until t , we just need to look at the derivative of $S(t)$ with respect to t . Since we are looking for a positive function and the survival probability diminished, we take the minus of the derivative:

$$\tilde{P}(t) = -\frac{dS}{dt} = -\frac{d}{dt} \operatorname{erf}\left(\frac{x_0}{\sqrt{4Dt}}\right) = -\frac{d}{dt}\left(\frac{x_0}{\sqrt{4Dt}}\right) \cdot \frac{d}{d\left(\frac{x_0}{\sqrt{4Dt}}\right)} \operatorname{erf}\left(\frac{x_0}{\sqrt{4Dt}}\right) \quad (13)$$

$$\boxed{\tilde{P}(t) = \frac{x_0}{\sqrt{4\pi Dt^3}} e^{-\frac{x_0^2}{4Dt}}} \quad (14)$$

Taking the mean:

$$\begin{aligned} \langle t \rangle &= \int_0^\infty t \tilde{P}(t) dt = \int_0^\infty \frac{x_0}{\sqrt{4\pi Dt}} e^{-\frac{x_0^2}{4Dt}} dt = \\ &= \left\{ \begin{array}{l} u = \frac{x_0}{\sqrt{4Dt}} \\ t = \frac{x_0^2}{4Du^2} \\ dt = -\frac{x_0^2}{2Du^3} du \end{array} \right\} = \\ &= -\int_\infty^0 \frac{u}{\sqrt{\pi}} e^{-u^2} \frac{x_0^2}{2Du^3} du = \frac{x_0^2}{2\sqrt{\pi}D} \int_0^\infty u^{-2} e^{-u^2} du \end{aligned} \quad (15)$$

Since for $u \sim 0$ this integrand diverges like u^{-2} , the integral doesn't converge and the mean first passage time is infinite.

$$\boxed{\langle t \rangle = \infty} \quad (16)$$

1.6 Discussion

Is it cheating to write this after we learned about heavytailed distributions? Well, it's too late for that.

Since for large times $\tilde{p}(t) \sim t^{-1.5}$, we can identify this with $\mu = 0.5$, which is indeed the Levy distribution we saw in class. The divergence of all moments is because all moments take the shape

$$\begin{aligned} M_n &= \frac{x_0}{\sqrt{4\pi D}} \int_0^\infty t^{n-1.5} e^{-\frac{x_0^2}{4Dt}} dt = \frac{x_0}{\sqrt{4\pi D}} \int_0^\infty \left(\frac{x_0^2}{4Du^2}\right)^{n-1.5} e^{-u^2} \frac{x_0^2}{2Du^3} du \propto \\ &\propto \int_0^\infty u^{-2n} e^{-u^2} du \end{aligned} \quad (17)$$

Which diverges for any $n \geq \frac{1}{2}$.

As we saw in class, the fact that the mean diverges will result in the cumulative FPT being dominated by a few rare events.

2 Markov processes and detailed balance

2.1 Master equation

The probability to be at a site p_i at time $t + 1$ is the sum of the probability to be at every site j that can transfer to site i at time t multiplied by the transition rate:

$$p_i(t + 1) = \sum_j p_j(t) k_{j,i} \quad (18)$$

Which, given nearest neighbors transition rates, is reduced to:

$$p_i(t + 1) = p_{i+1}(t) k_{i+1,i} + p_{i-1}(t) k_{i-1,i} + p_i(t) (1 - k_{i,i+1} - k_{i,i-1}) \quad (19)$$

But, this is a continoue Markov process, which means that time is not descrete. To get continous, time, we substract $p_i(t)$ from both sides and take the LHS to be a derivative to get the same time equation:

$$\boxed{\dot{p}_i = p_{i+1}(t) k_{i+1,i} + p_{i-1}(t) k_{i-1,i} - p_i(t) (k_{i,i+1} + k_{i,i-1})} \quad (20)$$

This means that the change in probability at every point is the difference between the incoming probability from the neighboring sites and the outgoing probability to those sites.

2.2 Steady states

As seen in the last equation, to get a steady state, where $\dot{p}_i = 0$ for each state, we require that:

$$p_{i+1} k_{i+1,i} + p_{i-1} k_{i-1,i} - p_i (k_{i,i+1} + k_{i,i-1}) = 0 \quad (21)$$

We can write this as:

$$p_{i+1} k_{i+1,i} = p_i (k_{i,i+1} + k_{i,i-1}) - p_{i-1} k_{i-1,i} \quad (22)$$

Renaing $i + 1 \rightarrow i$:

$$\boxed{p_i = \frac{k_{i-1,i} + k_{i-1,i-2}}{k_{i,i-1}} p_{i-1} - \frac{k_{i-2,i-1}}{k_{i,i-1}} p_{i-2}} \quad (23)$$

We also note the periodicity constraints:

$$\boxed{p_i = p_{i+n}} \quad (24)$$

and the noramlization constraint:

$$\boxed{\sum_{i=1}^n p_i = 1} \quad (25)$$

We can write the recursion relation by defining a matrix. To do so in a useful manner (to get all positive values), we will shift the index back and move p_i to the other side to get:

$$p_i(k_{i,i+1} + k_{i,i-1}) = p_{i+1}k_{i+1,i} + p_{i-1}k_{i-1,i} \quad (26)$$

$$p_i = \frac{k_{i+1,i}}{k_{i,i+1} + k_{i,i-1}}p_{i+1} + \frac{k_{i-1,i}}{k_{i,i+1} + k_{i,i-1}}p_{i-1} \quad (27)$$

Now it can be written in matrix form

$$M_{ij} = \begin{cases} \frac{k_{j,i}}{k_{i,j}+k_{i,j-2}} & j = i + 1 \\ \frac{k_{j,i}}{k_{i,j+2}+k_{i,j}} & j = i - 1 \\ 0 & else \end{cases} \quad (28)$$

The recursion relation then becomes an eigenvalue problem with eigenvalue 1:

$$A\vec{p} = \vec{p} \quad (29)$$

As we saw for the page rank in class, according to the Perron-Frobenius theorem, for a positive valued matrix there a unique solution and therefore a unique steady state.

2.3 Total probability current across all edges

The flux at each edge $i, i + 1$ is:

$$\boxed{\phi_{i,i+1} = p_i k_{i,i+1} - p_{i+1} k_{i+1,i}} \quad (30)$$

And for it to vanish at every edge, we require:

$$\boxed{p_{i+1} = \frac{k_{i,i+1}}{k_{i+1,i}}p_i} \quad (31)$$

2.4 Detailed balance

Let us set $k_{i,i\pm 1} = k_{\pm}$. This gives the matrix:

$$M_{ij} = \begin{pmatrix} 0 & \sigma_+ & 0 & \cdots \\ \sigma_- & 0 & \sigma_+ & \cdots \\ 0 & \sigma_- & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad \sigma_{\pm} = \frac{k_{\pm}}{k_+ + k_-} \quad (32)$$

It easy to see, from the symmetry of the system, that

$$v = \begin{pmatrix} 1 \\ 1 \\ 1 \\ \vdots \end{pmatrix} \quad (33)$$

is an eigenvector with eigenvalue $\lambda = \sigma_+ + \sigma_- = 1$, which is exactly the one we were looking for. Normalizing it, we get the steady state:

$$\boxed{p_i = \frac{1}{N}} \quad (34)$$

Since all probabilities are equal, the detailed balance condition becomes:

$$\boxed{k_+ = k_-} \quad (35)$$

2.5 Distinguishing steady state and detailed balance using the average occupation

From the previous example, it is clear that you cannot distinguish between a simple steady state and a detailed balance based solely on the average occupation. The detailed balance is a condition of transition rates between states, and thus even if the system is in a steady state, and the average occupation are all determined and constant, this doesn't mean the transition along every edge of the system will cancel out.

2.6 Kolmogorov Criterion

Let us assume steady state and prove Kolmogorov criterion. Let us define a reference state a and look at the path from it to site i and back. The path is

defined by all the sites in the middle, marked $s_1, \dots, s_n$. The product of transition rate from a to i and from i to a , along the path s , are:

$$\begin{aligned}\alpha_{a \rightarrow i, s} &= k_{a, s_1} k_{s_1, s_2} \cdots k_{s_{n-1}, s_n} k_{s_n, i} \\ \alpha_{i \rightarrow a, s} &= k_{i, s_n} k_{s_n, s_{n-1}} \cdots k_{s_2, s_1} k_{s_1, a}\end{aligned}\tag{36}$$

Dividing the two gives

$$\omega_i^{(a)} = \frac{k_{a, s_1} k_{s_1, s_2} \cdots k_{s_{n-1}, s_n} k_{s_n, i}}{k_{i, s_n} k_{s_n, s_{n-1}} \cdots k_{s_2, s_1} k_{s_1, a}}\tag{37}$$

Now, steady state means that

$$p_i k_{i, j} = p_j k_{j, i} \rightarrow \frac{k_{i, j}}{k_{j, i}} = \frac{p_j}{p_i}\tag{38}$$

So we can replace every matching pair in the nominator and denominator with the matching pair of probabilities:

$$\begin{aligned}\omega_i^{(a)} &= \frac{k_{a, s_1} k_{s_1, s_2} \cdots k_{s_{n-1}, s_n} k_{s_n, i}}{k_{i, s_n} k_{s_n, s_{n-1}} \cdots k_{s_2, s_1} k_{s_1, a}} = \\ &= \frac{k_{a, s_1}}{k_{s_1, a}} \cdot \frac{k_{s_1, s_2}}{k_{s_2, s_1}} \cdots \frac{k_{s_{n-1}, s_n}}{k_{s_n, s_{n-1}}} \cdot \frac{k_{s_n, i}}{k_{i, s_n}} = \\ &= \frac{p_{s_1}}{p_a} \cdot \frac{p_{s_2}}{p_{s_1}} \cdots \frac{p_{s_n}}{p_{s_{n-1}}} \cdot \frac{p_i}{p_{s_n}} = \frac{p_i}{p_a}\end{aligned}\tag{39}$$

So, the total ratio is independent of the path taken. More then that, selecting a closed loop by setting $i = a$ will result in $\omega_a^{(a)} = 1$, which means Kolmogorov criterion is met.

2.7 Steady state and detailed balance equivilance in tree graphs

A steady state is defined as:

$$\dot{p}_i = \sum_j (p_j k_{j, i} - p_i k_{i, j}) = 0 \quad \forall i\tag{40}$$

While detailed balance is defined as:

$$p_i k_{i, j} = p_j k_{j, i} \quad \forall i, j\tag{41}$$

It's clear to see, from summing of j in the detailed balance equation, that detailed balance requires a steady state. The other way around, however, is trickier. So, assuming a tree graph means that there are no loops. If there are no loops, the Kolmogorov criterion is met, and as we (didn't prove but we know it's true) saw in the last section, this means that there is detailed balance.

2.8 Adding energy

Adding energy and forcing the probability to act according to Boltzman distribution, will result in detailed balance:

$$\pi_i k_{i,j} = \pi_j k_{j,i} \quad (42)$$

looking like

$$e^{-\beta E_i} k_{i,j} = e^{-\beta E_j} k_{j,i} \quad (43)$$

$$\boxed{\frac{k_{i,j}}{k_{j,i}} = e^{-\beta(E_j - E_i)}} \quad (44)$$